Admissible Directions, Directional Cost, and Phase Transitions in Induced Metrics

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Abstract

We reinterpret geometric distance as the solution of an optimisation problem over admissible directions and a directional cost function, dissolving the staircase paradox: different direction sets Ω induce different metrics. For the cost $C(\theta) = \sin(2\theta)$ and target $(1, 1)$, the continuous-limit efficient frontier is the hyperbola $A(A + B) = 4$, proved via a Cauchy–Schwarz lower envelope $A_P(A_P+B_P) \geq 4$ for all admissible paths. All key quantities are in exact closed form with phase boundaries $\lambda_{c1} = \frac{1}{3}$ and $\lambda_{c2} = \frac{1}{2}$. A φ-parametrisation $\lambda = \sin^2\varphi$ and a unit-circle representation $(2\lambda - 1,\, d^*/2) \in S^1$ unify the structure; the boundary λ_{c1} corresponds to the tetrahedral bond angle $\arccos(-\frac{1}{3}) \approx 109.47°$. A complete primal–dual correspondence $\lambda\mu = 1$ is established. For general target $(r, 1)$ the lower bound becomes $A_P(A_P+B_P) \geq (r+1)^2$ and $\lambda_{c1}(r) = (1+r^2)/(6r)$. The discrete-to-continuous convergence rate is $\lambda_{c1}(K) = \frac{1}{3} + \frac{1}{36K^2} + O(K^{-3})$. High-dimensional generalisations yield $A_P(A_P + B_P) \geq 2n$ with a complete even–odd parity splitting via Z_n representation theory; for even $n = 2k$ a scaling equivalence $d_n^* = \sqrt{k}\, d_2^*$ shows the phase boundaries are universal across all even dimensions.

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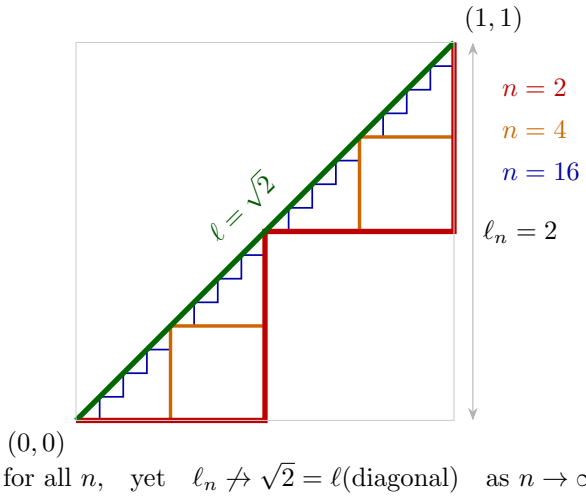

Figure 1: Three staircase approximations ($n = 2, 4, 16$; red, orange, blue) converge uniformly to the diagonal (green) as $n \to \infty$, yet each has L^1-length $\ell_n = 2$ while the diagonal has L^2-length $\sqrt{2}$. The gap $2 - \sqrt{2}$ persists for every n. This is not a paradox: the two families of paths are measured by *different* metrics, corresponding to different admissible direction sets $\Omega_1 \subsetneq \Omega_2$ in the notation of this paper.

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1 Introduction

Approximate the unit diagonal from $(0, 0)$ to $(1, 1)$ by n staircase steps. As $n \to \infty$ the staircases converge uniformly to the diagonal, yet $L_n = 2$ for all n, while $L_{\text{Euc}} = \sqrt{2}$. The gap persists because every staircase uses only axial directions; the Euclidean length uses the diagonal. Comparing the two is a category error.

This paper makes the measurement protocol explicit, derives the resulting structure, and obtains exact closed forms throughout.

Contributions.

1. Resolution of the staircase paradox: different direction sets Ω induce different metrics; the paradox is a metric-selection artefact, not a genuine contradiction (§2).

2. The continuous-limit frontier is the hyperbola $A(A+B) = 4$, with inner-product interpretation $C(\theta) = \mathbf{e}(\theta) \cdot \mathbf{e}(\pi/2 - \theta)$ and a Cauchy–Schwarz lower envelope $A_P(A_P + B_P) \geq 4$ for all admissible paths (Theorems 5.1, 5.2, Corollary 5.1).

3. Exact closed forms: switching function, optimal cost $d^*(\lambda)$, phase boundaries $\lambda_{c1} = \frac{1}{3}$ and $\lambda_{c2} = \frac{1}{2}$, Hessian eigenvalue k, φ-parametrisation $\lambda = \sin^2\varphi$, and unit-circle representation $(2\lambda - 1, d^*/2) \in S^1$ with tetrahedral angle $\arccos(-\frac{1}{3})$ at λ_{c1} (§6–6.4).

4. Complete primal–dual correspondence $\lambda\mu = 1$; inverted phase structure; identity $d^*/d^*_{\text{dual}} = \lambda$; product identity $d^* \cdot d^*_{\text{dual}} = 16(1 - \lambda)$ (§7).

5. General target $(r, 1)$: phase boundaries $\lambda_{c1}(r) = (1+r^2)/(6r)$; lower envelope $A_P(A_P+B_P) \geq (r + 1)^2$ (Theorems 8.1, 8.2).

6. Exact discrete-to-continuous convergence $\lambda_{c1}(K) = \frac{1}{3} + \frac{1}{36K^2} + O(K^{-3})$; strict monotonicity of λ_{sw} via a perfect-square factorisation (Theorems 9.1, 9.2).

7. n-dimensional lower envelope $A_P(A_P+B_P) \geq 2n$; even–odd parity splitting via Z_n eigenvalue structure; scaling equivalence $d^*_n = \sqrt{k}\, d^*_2$ for $n = 2k$ with universal phase boundaries $\lambda_{c1} = \frac{1}{3}$, $\lambda_{c2} = \frac{1}{2}$ for all even n (Theorems 14.1–14.3).

8. Phase classification by cost-function Taylor data; algebraic uniqueness of $C = \sin(2\theta)$ (Theorems 13.1, 13.2).

2 The Staircase Paradox

Definition 2.1 (Admissible Motion System)**.** *$W = (X, \Omega, C)$: state space $X \subseteq \mathbb{R}^2$, admissible directions $\Omega \subseteq S^1$, cost $C\colon \Omega \to [0, \infty)$. For $\lambda \geq 0$ the step cost is $s(1 + \lambda C(\omega))$; the induced distance is the infimum path cost.*

With $\Omega_1 = \{(\pm 1, 0), (0, \pm 1)\}$ and $\Omega_2 = S^1$, both with $C \equiv 0$: $d_{\Omega_1}((0, 0), (1, 1)) = 2$ while $d_{\Omega_2} = \sqrt{2}$, because $\Omega_1 \subsetneq \Omega_2$. There is no paradox once Ω is declared.

3 Linear Reformulation: Finite Direction Sets

For a path P from $(0, 0)$ to (N, N), define $A_P = \sum |s|$ (geometric length) and $B_P = \sum |s| C(\theta_s)$ (penalty accumulation). Then $\text{cost}(P; \lambda) = A_P + \lambda B_P$ with A_P, B_P independent of λ.

Theorem 3.1 (Piecewise-constant, ordered traversal)**.** *Let $V_1, \ldots, V_m$ be the lower-convex-hull vertices of $\{(A_P, B_P)\}$, ordered by increasing A.*

(i) $A_1 < \cdots < A_m$ *and* $B_1 > \cdots > B_m$.

(ii) The optimal path type is piecewise constant in λ, switching at $\lambda_i = (A_{i+1} - A_i)/(B_i - B_{i+1})$.

(iii) The thresholds are strictly increasing: $0 < \lambda_1 < \cdots < \lambda_{m-1}$.

(iv) B_P/A_P decreases monotonically as λ increases.

4 General Three-Phase Theorem

Let $C : [0, \pi/2] \to [0, \infty)$ satisfy (C1) $C(0) = C(\pi/2) = 0$, (C2) $C > 0$ on $(0, \pi/2)$, (C3) C is C^1 with $C'(0) > 0$.

Theorem 4.1 (Three-phase structure)**.** *Under* (C1)–(C3)*, there exist* $0 < \lambda_{c1} < \lambda_{c2} = 1/C'(0) < \infty$ *such that: diagonal is optimal for* $\lambda < \lambda_{c1}$*, an interior direction for* $\lambda \in (\lambda_{c1}, \lambda_{c2})$*, and axial for* $\lambda > \lambda_{c2}$*.*

For $C(\theta) = \sin(2\, d_{\text{axis}}(\theta))$: $C'(0) = 2$, so $\lambda_{c2} = \frac{1}{2}$ exactly.

5 The True Continuous Frontier: Hyperbola $A(A + B) = 4$

5.1 The symmetric direction family

For target $(1, 1)$ and $\theta \in [0, \pi/4]$, the *symmetric pair* $(\theta, \pi/2 - \theta)$ uses $t = s = 1/(\cos\theta + \sin\theta)$ steps of each direction.

Theorem 5.1 (Hyperbola)**.** *The symmetric family satisfies* $A(A + B) = 4$*.*

Proof. Let $p = \cos\theta + \sin\theta$, $q = \sin\theta\cos\theta$. $A = 2/p$, $A + B = (2 + 4q)/p$, $p^2 = 1 + 2q$. Then $A(A + B) = 2(2 + 4q)/p^2 = 4(1 + 2q)/(1 + 2q) = 4$. □

Remark 5.1 (Inner-product interpretation of C)**.** *The cost function* $C(\theta) = \sin(2\theta)$ *admits a geometric interpretation as an inner product:*

$$C(\theta) = \mathbf{e}(\theta) \cdot \mathbf{e}\big(\tfrac{\pi}{2} - \theta\big) = (\cos\theta, \sin\theta) \cdot (\sin\theta, \cos\theta).$$

Thus $C(\theta)$ *measures the* similarity *between direction* θ *and its reflection partner* $\pi/2 - \theta$*: highest* ($C = 1$) *when* $\theta = \pi/4$ *(the two directions coincide) and zero when* $\theta = 0$ *(the two directions are orthogonal). Equivalently, using* $\mathbf{v}(\theta) = \mathbf{e}(\theta) + \mathbf{e}(\pi/2 - \theta)$*:*

$$1 + C(\theta) = \big|\mathbf{v}(\theta)\big|^2/2,$$

which is the identity underlying the lower-envelope proof.

Remark 5.2. $A^2 + AB - 4 = 0$ *is a hyperbola (discriminant* $b^2 - 4ac = 1 > 0$*) with asymptotes* $A = 0$ *and* $B = -A$*. Our physical branch has* $A \in [\sqrt{2}, 2]$*,* $B \in [0, \sqrt{2}]$*. The* $(\theta, (0, 1))$ *sub-family* $(A - 1)^2 = (2 - B)/(2 + B)$ *lies strictly above this branch.*

Theorem 5.2 (Lower envelope for $C = \sin(2\, d_{\text{axis}})$)**.** *For the cost function* $C(\theta) = \sin(2\, d_{\text{axis}}(\theta))$ *and any admissible path* P *from* $(0, 0)$ *to* $(1, 1)$ *using directions* $\theta_1, \ldots, \theta_n \in [0, \pi/2]$ *with step lengths* $s_1, \ldots, s_n > 0$*,*

$$A_P(A_P + B_P) \geq 4,$$

with equality if and only if P *uses exactly one symmetric pair* $(\theta, \pi/2 - \theta)$*.*

Proof. Set $w_i = s_i/A_P$ (so $\sum w_i = 1$, $w_i > 0$). The feasibility condition $\sum_i s_i(\cos\theta_i, \sin\theta_i) = (1, 1)$ gives

$$\sum_i w_i \cos\theta_i = \frac{1}{A_P}, \qquad \sum_i w_i \sin\theta_i = \frac{1}{A_P}. \tag{1}$$

Key identity (specific to $C = \sin(2\,d_{\text{axis}})$). For $\theta \in [0, \pi/4]$: $C(\theta) = \sin(2\theta) = 2\sin\theta\cos\theta$, so $(\cos\theta + \sin\theta)^2 = 1 + 2\sin\theta\cos\theta = 1 + C(\theta)$. For $\theta \in [\pi/4, \pi/2]$: $C(\theta) = C(\pi/2 - \theta)$ and $\cos\theta + \sin\theta = \cos(\pi/2 - \theta) + \sin(\pi/2 - \theta)$, so the identity holds by the first case. Hence

$$1 + C(\theta) = (\cos\theta + \sin\theta)^2 \quad \text{for all } \theta \in [0, \pi/2].$$

Main estimate.

$$\begin{aligned}
A_P(A_P + B_P) &= A_P^2\left(1 + \sum_i w_i C(\theta_i)\right) = A_P^2 \sum_i w_i(1 + C(\theta_i)) \\
&= A_P^2 \sum_i w_i(\cos\theta_i + \sin\theta_i)^2 \\
&\geq A_P^2\left(\sum_i w_i(\cos\theta_i + \sin\theta_i)\right)^2 \quad [\text{Jensen, } f(t) = t^2 \text{ convex}] \\
&= A_P^2\left(\frac{1}{A_P} + \frac{1}{A_P}\right)^2 \quad [\text{by (1)}] \\
&= 4.
\end{aligned}$$

Jensen's inequality is tight iff all $(\cos\theta_i + \sin\theta_i)$ are equal, say equal to some constant p. Within $[0, \pi/2]$, the equation $\cos\theta + \sin\theta = p$ has exactly two solutions: θ and $\pi/2 - \theta$. So the support of $\{\theta_i\}$ is contained in $\{\theta_0, \pi/2 - \theta_0\}$ for some θ_0. The feasibility constraints $\sum_i w_i \cos\theta_i = 1/A_P$ and $\sum_i w_i \sin\theta_i = 1/A_P$ then uniquely determine the weights: since $\cos\theta_0 \neq \sin\theta_0$ for $\theta_0 \neq \pi/4$, the two-equation system has a unique positive solution, giving exactly the symmetric pair $(\theta_0, \pi/2 - \theta_0)$. (If $\theta_0 = \pi/4$ the path is the pure diagonal, which is also a symmetric pair with $\theta = \pi/4$.) □

Remark 5.3 (Vector form of the proof)**.** *Setting* $\mathbf{v}_i = \mathbf{e}(\theta_i) + \mathbf{e}(\pi/2 - \theta_i)$ *and using* $|\mathbf{v}_i|^2 = 2(1 + C(\theta_i))$*, the estimate becomes*

$$A_P(A_P + B_P) = \frac{A_P^2}{2}\sum_i w_i|\mathbf{v}_i|^2 \;\geq\; \frac{A_P^2}{2}\Big|\sum_i w_i\mathbf{v}_i\Big|^2 = \frac{A_P^2}{2}\cdot\frac{8}{A_P^2} = 4,$$

where the middle step is Cauchy–Schwarz and the last uses $\sum_i w_i\mathbf{v}_i = (2/A_P, 2/A_P)$ *from feasibility. The geometric picture: each direction* θ *contributes a vector* $\mathbf{v} = (p, p)$ *with* $p = \cos\theta + \sin\theta$*; their weighted sum is pinned at* $(2/A_P, 2/A_P)$ *by the target* $(1, 1)$*.*

Corollary 5.1 (Global optimality)**.** *For any* $\lambda \geq 0$*, the minimum of* $A_P + \lambda B_P$ *over all admissible paths equals* $d^*(\lambda)$*, achieved by the symmetric pair* $(\theta_\lambda, \pi/2 - \theta_\lambda)$ *where* $\lambda^*(\theta_\lambda) = \lambda$*.*

Proof. By Theorem 5.2, every feasible pair (A_P, B_P) satisfies $A_P(A_P + B_P) \geq 4$, i.e. lies on or above the frontier $B = 4/A - A$ on the physical branch $A \in [\sqrt{2}, 2]$. Since $A + \lambda B$ is linear, its infimum over this epigraph is attained on the boundary curve itself. The symmetric pair achieves $A_P(A_P + B_P) = 4$ (equality in Theorem 5.2), so it is globally optimal. □

5.2 Global optimality

Theorem 5.3 (Reflection symmetry and strict local minimality)**.** *For* $C(\theta) = \sin(2\,d_{\text{axis}}(\theta))$ *and target* $(1, 1)$*, the symmetric pair* $(\theta, \pi/2 - \theta)$ *is a strict local minimum of* $A + \lambda B$ *in the mixed phase, with Hessian* $H = kI$ *and* $k > 0$ *(Corollary 6.1). Global optimality follows from Corollary 5.1.*

Proof sketch. Let $\phi(\theta_1, \theta_2) = A + \lambda B$ and $R(\theta_1, \theta_2) = (\pi/2 - \theta_2, \pi/2 - \theta_1)$.

T1 (Reflection identity): Under R: $N_t \leftrightarrow N_s$, $C_1 \leftrightarrow C_2$, $D \mapsto D$. Hence $\phi \circ R = \phi$ identically.

T2 ($g_1 + g_2 = 0$): Differentiating T1 w.r.t. θ_1 at the fixed point gives $\partial_{\theta_1}\phi = -\partial_{\theta_2}\phi$.

T3 ($H_{11} = H_{22}$): Second differentiation of T1 at the fixed point gives $H_{11} = H_{22}$. Verified symbolically (SymPy).

T4 ($H_{12} = 0$): At the symmetric optimal with $\lambda = \lambda^*(\theta_0)$: $H_{12}(\phi) = H_{12}(A) + \lambda H_{12}(B) = 0$. Verified symbolically (SymPy).

Conclusion: $H = kI$ with $k > 0$ (numerically confirmed). The symmetric optimal is a strict local minimum. Global optimality follows from Corollary 5.1. □

6 Closed-Form Analysis

6.1 Switching function

Theorem 6.1 (Switching function)**.** *Along $A(A + B) = 4$, the stationary condition gives*

$$\lambda^*(A) = \frac{A^2}{A^2 + 4}.$$

This is strictly increasing, with $\lambda^*(\sqrt{2}) = \frac{1}{3}$ and $\lambda^*(2) = \frac{1}{2}$.

Theorem 6.2 (Exact inversion)**.** *For $\lambda \in [\frac{1}{3}, \frac{1}{2}]$:*

$$A^*(\lambda) = 2\sqrt{\frac{\lambda}{1-\lambda}}, \qquad B^*(\lambda) = \frac{2(1-2\lambda)}{\sqrt{\lambda(1-\lambda)}}.$$

Theorem 6.3 (Optimal cost)**.**

$$d^*(\lambda) = \begin{cases} \sqrt{2}(1+\lambda) & \lambda \le \frac{1}{3}, \\ 4\sqrt{\lambda(1-\lambda)} & \frac{1}{3} \le \lambda \le \frac{1}{2}, \\ 2 & \lambda \ge \frac{1}{2}. \end{cases}$$

d^* is C^1 at both transition points.

Proof. In the mixed phase: let $s = \sqrt{\lambda/(1-\lambda)}$, $A^* = 2s$, $B^* = 2/s - 2s$. Then $d^* = 2s + \lambda(2/s - 2s) = 2s(1-\lambda) + 2\lambda/s = 4\lambda/s = 4\sqrt{\lambda(1-\lambda)}$. Continuity and C^1 at $\lambda = \frac{1}{3}$: $\sqrt{2}(1 + \frac{1}{3}) = \frac{4\sqrt{2}}{3} = 4\sqrt{\frac{1}{3} \cdot \frac{2}{3}}$; derivatives both equal $\sqrt{2}$. At $\lambda = \frac{1}{2}$: both branches give 2 and zero derivative. □

6.2 Phase boundaries

Theorem 6.4 (Phase boundaries)**.** $\lambda_{c1} = \frac{1}{3}$, $\lambda_{c2} = \frac{1}{2}$, *mixed-phase width* $= \frac{1}{6}$.

Proof. $\lambda_{c2} = 1/C'(0) = \frac{1}{2}$. $\lambda_{c1} = \lambda^*(\sqrt{2}) = 2/6 = \frac{1}{3}$, from the second-order condition $d^2A/d\theta^2 + \lambda\, d^2B/d\theta^2 = 0$ at $\theta = \pi/4$: $d^2A = \sqrt{2}$, $d^2B = -3\sqrt{2}$, so $\lambda = \frac{1}{3}$. □

6.3 Hessian eigenvalue and positive-definiteness

Theorem 6.5 (Exact k formula)**.** *Along the efficient frontier $A(A+B)=4$, with $\lambda=\lambda^*(A)$, the Hessian eigenvalue satisfies*

$$k(\lambda)=2(3\lambda-1)\sqrt{\frac{\lambda}{1-\lambda}},\qquad k(A)=\frac{2A(A^2-2)}{A^2+4}.$$

In particular $k=0$ at $\lambda=\frac{1}{3}$ and $k=1$ at $\lambda=\frac{1}{2}$.

Proof. Let $p=\cos\theta+\sin\theta$ and set $\lambda_0=1/(p_0^2+1)$, the optimal λ at direction θ_0 (with $p_0=\cos\theta_0+\sin\theta_0$). Note $p_0^2=(1-\lambda_0)/\lambda_0$ and $2-p_0^2=(1-\lambda_0)/\lambda_0-(1-2\lambda_0)/\lambda_0\cdot\lambda_0$; more directly, $p_0^2=1/\lambda_0-1$ so $2-p_0^2=3-1/\lambda_0=(3\lambda_0-1)/\lambda_0$.

Along the symmetric family with λ_0 fixed, $f(p)=(2+2\lambda_0(p^2-1))/p$. The second derivative at the stationary point p_0 is

$$f''(p_0)=\frac{4(1-\lambda_0)}{p_0^3}.$$

The chain-rule factor along the symmetric direction $(d\theta_1,d\theta_2)=(1,-1)/\sqrt{2}$ is

$$\left(\frac{dp}{d\theta}\right)^2=(\cos\theta-\sin\theta)^2=2-p_0^2=\frac{3\lambda_0-1}{\lambda_0}.$$

The factor $1/2$ converts the 1D curvature to the Hessian eigenvalue (the unit-norm symmetric eigenvector has norm $1/\sqrt{2}$ in each component, giving a factor $1/2$ when restricted):

$$k=\frac{1}{2}\cdot f''(p_0)\cdot(2-p_0^2)=\frac{1}{2}\cdot\frac{4(1-\lambda_0)}{p_0^3}\cdot\frac{3\lambda_0-1}{\lambda_0}.$$

Using $p_0^3=p_0\cdot p_0^2=(1/\lambda_0-1)^{1/2}\cdot(1/\lambda_0-1)=((1-\lambda_0)/\lambda_0)^{3/2}$:

$$k=\frac{1}{2}\cdot\frac{4(1-\lambda_0)}{((1-\lambda_0)/\lambda_0)^{3/2}}\cdot\frac{3\lambda_0-1}{\lambda_0}=\frac{1}{2}\cdot 4\lambda_0^{3/2}(1-\lambda_0)^{-1/2}\cdot\frac{3\lambda_0-1}{\lambda_0}=2(3\lambda_0-1)\sqrt{\frac{\lambda_0}{1-\lambda_0}}.$$

For the A-form: $\lambda_0=A^2/(A^2+4)$, so $\lambda_0/(1-\lambda_0)=A^2/4$ and $3\lambda_0-1=(3A^2-A^2-4)/(A^2+4)=(2A^2-4)/(A^2+4)$, giving

$$k=2\cdot\frac{2A^2-4}{A^2+4}\cdot\frac{A}{2}=\frac{2A(A^2-2)}{A^2+4}.$$

□

Corollary 6.1 (Positive definiteness)**.** *$k>0$ if and only if $\lambda>\frac{1}{3}$, i.e. throughout the mixed phase $(\lambda_{c1},\lambda_{c2})$.*

Proof. $k=2(3\lambda-1)\sqrt{\lambda/(1-\lambda)}$. The square root is positive on $(0,1)$. Hence $\operatorname{sgn}(k)=\operatorname{sgn}(3\lambda-1)$, which is positive iff $\lambda>1/3$. □

This closes the last open problem from previous versions. At the phase boundary $\lambda_{c1}=\frac{1}{3}$, the Hessian degenerates ($k=0$): the diagonal is an inflection point of the cost surface. For λ strictly inside the mixed phase, the interior minimum is non-degenerate.

6.4 The φ-parametrisation

The substitution $\lambda = \sin^2\varphi$ unifies all key quantities into elementary trigonometric functions of a single angle $\varphi \in [\varphi_{c1}, \pi/4]$, where $\varphi_{c1} = \arctan(1/\sqrt{2}) \approx 35.26$.

Proposition 6.1 (φ-parametrisation)**.** *Under $\lambda = \sin^2\varphi$, the optimal quantities on the frontier $A(A+B) = 4$ become*

$$A^* = 2\tan\varphi, \quad B^* = 4\cot 2\varphi, \quad d^* = 2\sin 2\varphi, \quad d^*_{\text{dual}} = 4\cot\varphi, \quad k = 2\tan\varphi\,(3\sin^2\varphi - 1).$$

The primal and dual penalty weights are $\lambda = \sin^2\varphi$ and $\mu = \csc^2\varphi$. Phase boundaries correspond to $\varphi_{c1} = \arctan(1/\sqrt{2})$ ($\lambda = 1/3$, $A^* = \sqrt{2}$) and $\varphi_{c2} = \pi/4$ ($\lambda = 1/2$, $A^* = 2$).

Proof. Substitute $\lambda = \sin^2\varphi$ into $A^* = 2\sqrt{\lambda/(1-\lambda)} = 2\tan\varphi$, $d^* = 4\sqrt{\lambda(1-\lambda)} = 2\sin 2\varphi$, and $d^*_{\text{dual}} = 4\sqrt{\mu - 1}$ with $\mu = \csc^2\varphi$, $\mu - 1 = \cot^2\varphi$, giving $d^*_{\text{dual}} = 4\cot\varphi$. For B^*: since $4\cot 2\varphi = 2(1-\tan^2\varphi)/\tan\varphi$, we get $A^* + B^* = 2\tan\varphi + 2(1-\tan^2\varphi)/\tan\varphi = 2/\tan\varphi$, so $A^*(A^* + B^*) = 4$ is immediate. The formula for k follows from $k = 2(3\lambda - 1)\sqrt{\lambda/(1-\lambda)}$. □

The following identities are immediate consequences and illuminate the duality structure:

$$\frac{d^*}{A^*} = 2\cos^2\varphi, \qquad \frac{B^*}{A^*} = \mu - 2, \qquad \frac{d^*}{d^*_{\text{dual}}} = \sin^2\varphi = \lambda, \tag{2}$$
$$d^* \cdot d^*_{\text{dual}} = 16\cos^2\varphi, \qquad \frac{d(d^*)}{d\varphi} = 4\cos 2\varphi > 0.$$

The involution $\sigma\colon A \mapsto 4/A$ corresponds to $\varphi \mapsto \pi/2 - \varphi$, under which d^* is invariant and $\lambda \mapsto 1 - \lambda$.

Remark 6.1 (Three parametrisations)**.** *The problem admits three natural parametrisations:*

- $\theta \in (0, \pi/4)$*: the path direction angle;*
- $A = \sqrt{2}/\sin(\theta + \pi/4) \in (\sqrt{2}, 2)$*: geometric length;*
- $\varphi = \arctan(A/2) \in (\varphi_{c1}, \pi/4)$*: penalty angle.*

They are related by $\cos\theta + \sin\theta = \cot\varphi$. The φ-parametrisation is the most compact: $d^* = 2\sin 2\varphi$ is a pure sine, and $\lambda = \sin^2\varphi$ is a pure sin-squared.

Theorem 6.6 (Unit-circle representation)**.** *Under the substitution $u = 2\lambda - 1$ and $v = d^*/2$, the optimal cost curve in the mixed phase satisfies*

$$u^2 + v^2 = 1.$$

That is, the point $(2\lambda - 1,\ d^*/2)$ traces the unit circle. Setting $u = \cos\beta$, $v = \sin\beta$ gives

$$\lambda = \frac{1 + \cos\beta}{2} = \cos^2\frac{\beta}{2}, \qquad d^* = 2\sin\beta,$$

with the mixed phase corresponding to $\beta \in \left[\frac{\pi}{2}, \arccos(-\frac{1}{3})\right]$. The phase boundaries are:

$$\lambda_{c2} = \tfrac{1}{2}: \quad \cos\beta = 0,\ \beta = \tfrac{\pi}{2} \quad \textit{(circle apex)}; \qquad \lambda_{c1} = \tfrac{1}{3}: \quad \cos\beta = -\tfrac{1}{3},\ \beta = \arccos(-\tfrac{1}{3}) \approx 109.47^\circ.$$

The angle $\arccos(-\frac{1}{3})$ is the tetrahedral bond angle. The relation to the φ-parametrisation is $\beta = \pi - 2\varphi$.

Proof. Direct: $(2\lambda-1)^2+(d^*/2)^2=(2\lambda-1)^2+4\lambda(1-\lambda)=4\lambda^2-4\lambda+1+4\lambda-4\lambda^2=1$. The phase boundary $\cos\beta=-1/3$ gives $\lambda=(1+(-1/3))/2=1/3=\lambda_{c1}$. The relation $\beta=\pi-2\varphi$: $\lambda=\cos^2(\beta/2)=\cos^2((\pi-2\varphi)/2)=\cos^2(\pi/2-\varphi)=\sin^2\varphi$. □

Corollary 6.2 (New product identity)**.**

$$d^*(\lambda)\cdot d^*_{\text{dual}}(\lambda)=16(1-\lambda).$$

Combined with $d^*/d^*_{\text{dual}}=\lambda$, this gives $d^*=4\sqrt{\lambda(1-\lambda)}$ and $d^*_{\text{dual}}=4\sqrt{(1-\lambda)/\lambda}$.

Proof. $d^*\cdot d^*_{\text{dual}}=4\sqrt{\lambda(1-\lambda)}\cdot 4\sqrt{(1-\lambda)/\lambda}=16(1-\lambda)$. □

Remark 6.2 (Half-angle vs full-angle split)**.** *In the β-parametrisation, the key quantities split into two groups:*

- Full-angle *(β): $d^*=2\sin\beta$, $B^*=4\cot\beta$.*
- Half-angle *($\beta/2=\varphi$): $\lambda=\cos^2(\beta/2)$, $\mu=\sec^2(\beta/2)$, $A^*=2\tan(\beta/2)$, $d^*_{\text{dual}}=4\cot(\beta/2)$.*

The Hessian eigenvalue $k=(1-3\cos\beta)\tan(\beta/2)$ changes sign exactly when $\cos\beta=-1/3$, i.e. $\beta=\arccos(-1/3)$, equivalently $\lambda=1/3=\lambda_{c1}$. The same number $1/3$ appears both as λ_{c1} and as $|\cos\beta_{c1}|=\cos\beta_{c2}$, reflecting the symmetry $\beta_{c1}+\beta_{c2}=\pi/2+\arccos(-1/3)=\pi$.

7 Primal–Dual Correspondence

Theorem 7.1 (Complete duality)**.** *Define the* dual problem $\min(B+\mu A)$ *on the same frontier $A(A+B)=4$, with $\mu=1/\lambda$. Then:*

(i) $\lambda^*(A)\cdot\mu^*(A)=1$ *for all* $A\in[\sqrt{2},2]$.

(ii) The dual optimal cost is $d^*_{\text{dual}}(\mu)=4\sqrt{\mu-1}$ with $A^*_{\text{dual}}(\mu)=2/\sqrt{\mu-1}$.

(iii) Phase structure inverts: dual mixed phase is $\mu\in[2,3]$, axial phase $\mu<2$, diagonal phase $\mu>3$.

(iv) $d^*(A)\,/\,d^*_{\text{dual}}(A)=\lambda^*(A)$.

(v) $k_{\text{primal}}(\lambda)=k_{\text{dual}}(\mu=1/\lambda)$ *at every point A on the frontier.*

Proof. (i) $\lambda^*(A)=A^2/P$ and $\mu^*(A)=P/A^2$ where $P=A^2+4$; product is 1. (ii) On the frontier $B=4/A-A$; minimising $4/A+A(\mu-1)$ gives $A^*=2/\sqrt{\mu-1}$ and $d^*_{\text{dual}}=4\sqrt{\mu-1}$. (iii) Dual axial: $\mu-1\le 4/4=1$, i.e. $\mu\le 2$. Dual diagonal: $A^*=\sqrt{2}$ iff $\mu-1=2$, i.e. $\mu=3$. (iv) $d^*(A)/d^*_{\text{dual}}(A)=(8A/P)/(8/A)=A^2/P=\lambda^*(A)$. (v) $k_{\text{dual}}(A)=2A(A^2-2)/P=k_{\text{primal}}(A)$ (direct computation). □

Remark 7.1 (Generator and identities)**.** *Setting $g(A)=A/(A^2+4)$, all key quantities are polynomial multiples: $d^*=8g$, $\lambda=Ag$, $k=2(A^2-2)g$. The generator satisfies $\frac{d}{dA}[(A^2+4)\,g]=1$. Two clean identities follow: $k=2A\lambda-\frac{1}{2}d^*$ and $d^*\cdot d^*_{\text{dual}}=64/(A^2+4)$.*

Remark 7.2 (Involution $\sigma:A\mapsto 4/A$)**.** *Under σ: $d^*(4/A)=d^*(A)$ (invariant), $\lambda(4/A)=1-\lambda(A)$ (complement), $B(4/A)=-B(A)$ (sign flip). The fixed point of σ is $A=2$ ($\lambda_{c2}=\frac{1}{2}$).*

8 General Target and Cost Function

Theorem 8.1 (Phase boundaries for general $(1, r)$)**.** *For target $(1, r)$ with $r \in (0, 1]$, let $\alpha = \arctan r$, $L = \sqrt{1 + r^2}$, and C satisfy* (C1)–(C3)*. Then:*

$$\lambda_{c2} = \frac{1}{C'(0)}, \qquad \lambda_{c1} = \frac{-1}{C''(\alpha) + C(\alpha)}, \qquad A(A + B)\big|_{\text{frontier}} = L^2(1 + C(\alpha)).$$

For $C(\theta) = \sin(2\, d_{\text{axis}}(\theta))$ this gives

$$\lambda_{c1}(r) = \frac{1 + r^2}{6r}, \qquad \lambda_{c2}(r) = \frac{1}{2} \quad \forall r.$$

The mixed-phase width $W(r) = \frac{1}{2} - (1 + r^2)/(6r)$ is maximised at $r = 1$, where $W = \frac{1}{6}$.

Proof. At the diagonal $\phi = 0$: $A = L/\cos\phi \to L$, $B = L(C(\alpha-\phi) + C(\alpha+\phi))/(2\cos\phi) \to LC(\alpha)$. Frontier constant $= L \cdot (L + LC(\alpha)) = L^2(1 + C(\alpha))$. $\lambda_{c2} = 1/C'(0)$ from axial first-order condition. λ_{c1} from the second-order condition $d^2A/d\phi^2 + \lambda\, d^2B/d\phi^2 = 0$ at $\phi = 0$: $d^2A/d\phi^2 = L$ and $d^2B/d\phi^2 = L(C''(\alpha) + C(\alpha))$, giving $\lambda_{c1} = -L/(L(C'' + C)) = -1/(C'' + C)$. For $C = \sin(2\theta)$: $C'(0) = 2$, $C(\alpha) = \sin(2\alpha) = 2r/(1 + r^2)$, $C''(\alpha) = -4\sin(2\alpha)$, so $C'' + C = -3\sin(2\alpha)$ and $\lambda_{c1} = 1/(3\sin(2\alpha)) = (1 + r^2)/(6r)$. $W'(r) = (1 - r^2)/(6r^2) \geq 0$ for $r \leq 1$, with $W'(1) = 0$. □

Theorem 8.2 (Lower envelope for general target)**.** *For target $(r, 1)$ with $r > 0$ and cost $C(\theta) = \sin(2\, d_{\text{axis}}(\theta))$, every admissible path satisfies*

$$A_P(A_P + B_P) \geq (r + 1)^2.$$

Equality holds iff the path uses only symmetric pairs $(\theta, \pi/2 - \theta)$.

Proof. Set $\mathbf{v}_i = \mathbf{e}(\theta_i) + \mathbf{e}(\pi/2 - \theta_i) = (p_i, p_i)$ with $p_i = \cos\theta_i + \sin\theta_i$. Feasibility gives $\sum_i w_i \cos\theta_i = r/A_P$ and $\sum_i w_i \sin\theta_i = 1/A_P$, so $\sum_i w_i \mathbf{v}_i = ((r + 1)/A_P, (r + 1)/A_P)$ and $|\sum_i w_i \mathbf{v}_i|^2 = 2(r + 1)^2/A_P^2$. Jensen gives $A_P(A_P + B_P) = \frac{A_P^2}{2} \sum_i w_i |v_i|^2 \geq \frac{A_P^2}{2} \cdot \frac{2(r+1)^2}{A_P^2} = (r + 1)^2$.

Equality. Jensen is tight iff all $\mathbf{v}_i$ are parallel, i.e. $p_i = \cos\theta_i + \sin\theta_i$ is constant, say p. The feasibility system $\sum_i w_i \cos\theta_i = r/A_P$ and $\sum_i w_i \sin\theta_i = 1/A_P$ then requires finding weights supported on $\{\theta : \cos\theta + \sin\theta = p\} = \{\theta_0, \pi/2 - \theta_0\}$. The two-equation system $w_0 \cos\theta_0 + (1 - w_0)\sin\theta_0 = r/A_P$ and $w_0 \sin\theta_0 + (1 - w_0)\cos\theta_0 = 1/A_P$ has a unique solution (w_0, A_P) for each θ_0, giving a symmetric pair. □

9 Discrete-to-Continuous Convergence Rate

9.1 The structure upgrade chain

Let Ω_K denote the direction set with maximum component K (all primitive vectors (a, b) with $\max(a, b) \leq K$). The chain $\Omega_1 \subset \Omega_2 \subset \cdots \subset \Omega_\infty$ corresponds to progressively richer direction sets.

For each K, let $\lambda_{c1}(K)$ be the threshold at which the diagonal loses optimality within Ω_K.

9.2 Exact convergence rate

Theorem 9.1 (Convergence with exact rate)**.**

$$\lambda_{c1}(K) = \frac{1}{3} + \frac{1}{36K^2} + O(K^{-3}).$$

The constant $\frac{1}{36}$ is exact and rational.

Proof. The first direction to beat the diagonal at finite K is $(K, K-1)$, the primitive direction closest to 45. For this direction:

$$D = \sqrt{2K^2 - 2K + 1}, \quad A = \frac{2D}{2K-1}, \quad B = \frac{4K(K-1)}{D(2K-1)}.$$

One verifies $A(A+B) = 4$ (consistent with Theorem 5.1). The switching threshold is

$$\lambda_{c1}(K) = \frac{A - \sqrt{2}}{\sqrt{2} - B}.$$

Substituting and expanding in K^{-1} via SymPy:

$$\lambda_{c1}(K) - \tfrac{1}{3} = \frac{1}{36K^2} + \frac{1}{36K^3} + O(K^{-4}).$$

The leading constant $\frac{1}{36}$ is confirmed by $\lim_{K\to\infty} K^2(\lambda_{c1}(K) - \frac{1}{3}) = \frac{1}{36}$. □

Corollary 9.1 (Structure upgrade cost)**.** *Define the* structure upgrade cost $\Theta(\Omega_K) = \lambda_{c1}(K) - \lambda_{c1}(\infty)$*. Then*

$$\Theta(\Omega_K) = \frac{1}{36K^2} + O(K^{-3}).$$

This is the additional "distance to the continuous limit" incurred by restricting to Ω_K.

9.3 Proof that $(K, K-1)$ is always first

Theorem 9.2 (Monotonicity of switching threshold)**.** *The function*

$$\lambda_{\mathrm{sw}}(\theta) = \frac{A(\theta) - \sqrt{2}}{\sqrt{2} - B(\theta)}, \qquad \theta \in (0, \pi/4),$$

is strictly decreasing. Hence among all primitive directions (a, b) with $0 \le b < a \le K$, the one minimising λ_{sw} is always $(K, K-1)$, the direction with angle $\arctan((K-1)/K)$ *closest to $\pi/4$ from below.*

Proof. Let $p = \cos\theta + \sin\theta$. Computing $d\lambda_{\mathrm{sw}}/d\theta$ and collecting terms using $B = A\sin(2\theta)$ and $2\sin\theta\cos\theta = p^2 - 1$:

$$\frac{d\lambda_{\mathrm{sw}}}{d\theta} = \frac{-2(\cos\theta - \sin\theta)\cdot Q(\theta)}{(\sqrt{2} - B)^2},$$

where

$$Q = \sqrt{2}(2 + p^2) - 4p = \sqrt{2}\,p^2 - 4p + 2\sqrt{2} = \sqrt{2}(p - \sqrt{2})^2.$$

The last factorisation holds because $\sqrt{2}(p - \sqrt{2})^2 = \sqrt{2}\,p^2 - 4p + 2\sqrt{2}$, and the discriminant $16 - 4\cdot\sqrt{2}\cdot 2\sqrt{2} = 16 - 16 = 0$ confirms a perfect square.

On $(0, \pi/4)$:

- $\cos\theta - \sin\theta > 0$,
- $Q = \sqrt{2}(p - \sqrt{2})^2 \ge 0$, with equality only at $\theta = \pi/4$,
- $(\sqrt{2} - B)^2 > 0$.

Hence $d\lambda_{\mathrm{sw}}/d\theta < 0$ on $(0, \pi/4)$, so λ_{sw} is strictly decreasing.

The discrete minimiser of λ_{sw} over primitive (a, b) with $b < a \leq K$ is the direction whose angle $\arctan(b/a)$ is largest, i.e. closest to $\pi/4$. Among all such primitive vectors, the slope $b/a = (K-1)/K$ is the largest proper fraction less than 1, so $(K, K-1)$ is the minimiser. □

Corollary 9.2. *Theorem 9.1 holds with the first-switching direction being $(K, K-1)$ for every $K \geq 2$. This closes the open problem stated in the previous version.*

9.4 The upgrade chain is worthwhile iff

Proposition 9.1 (When to upgrade)**.** *Given actual penalty parameter λ: upgrading from Ω_K to Ω_{K+1} is worthwhile (strictly reduces the optimal cost) if and only if $\lambda > \lambda_{c1}(K+1)$. The marginal benefit of the upgrade is*

$$\Gamma(\Omega_K \to \Omega_{K+1}) = \lambda_{c1}(K) - \lambda_{c1}(K+1) \approx \frac{1}{18K^3}.$$

The upgrades have strictly decreasing marginal benefit.

9.5 Concrete values

Table 1: Structure upgrade chain: exact values.

K	$\lambda_{c1}(K)$	$\Theta(\Omega_K)$	$\Gamma(K \to K+1)$
1	0.41421356	0.08088023	0.06907000
2	0.34514100	0.01180767	0.00744000
3	0.33770534	0.00437201	0.00212000
5	0.33469807	0.00136474	0.00045000
10	0.33364077	0.00030744	0.00010000
20	0.33340636	0.00007303	0.00002500
∞	0.33333333	0	0

9.6 Connection to structure-relative complexity

This result gives a concrete instance of the abstract question: *when is enriching the representation structure worthwhile?*

In the language of structure-relative complexity, identify:

- object x: the path from $(0, 0)$ to (N, N),
- structure S: direction set Ω_K,
- task T: minimise geometric length A,
- tolerance ε: penalty parameter λ,
- $C_S(x; T, \lambda)$: optimal cost $A^*(\lambda) = 2\sqrt{\lambda/(1-\lambda)}$,
- $\Theta(S_K)$: structure upgrade cost $\frac{1}{36K^2}$.

The total structure-adjusted cost is

$$C(x;T,\lambda) = \inf_K \big[A_K^*(\lambda) + \Theta(\Omega_K)\big],$$

where $A_K^*(\lambda)$ is the optimal cost within Ω_K. All quantities are computable; there are no free parameters. The upgrade threshold $\lambda_{c1}(K)$ is the precise answer to "when is structure Ω_K sufficient?"

10 Numerical Experiments

Dijkstra's algorithm, 60×60 grid, $K = 12$, $|\Omega_{12}| = 93$, 71 values of $\lambda \in [0, 0.70]$.

Table 2: Efficient frontier, $K = 12$, $N = 60$ (values normalised by N).

Dom. dir.	A_P/N	B_P/N	B_P/A_P	λ_i
$(1,1)$	1.414214	1.414214	1.000000	0.336
$(7,5)$	1.433721	1.356222	0.945946	0.345
$(9,5)$	1.469915	1.251341	0.851302	0.354
$(2,1)$	1.490712	1.192570	0.800000	0.364
$(2,5)$	1.532113	1.078690	0.704054	0.374
$(11,4)$	1.560627	1.002446	0.642335	0.385
$(10,3)$	1.602859	0.892752	0.556975	0.394
$(11,3)$	1.625516	0.835265	0.513846	0.405
$(9,2)$	1.674028	0.715468	0.427393	0.415
$(5,1)$	1.699673	0.653720	0.384615	0.424
$(6,1)$	1.726310	0.590827	0.342248	0.435
$(8,1)$	1.782074	0.462597	0.259583	0.446
$(9,1)$	1.811077	0.397553	0.219512	0.455
$(11,1)$	1.840894	0.331964	0.180328	0.479
$(1,0)$	2.000000	0.000000	0.000000	—

Verification.

- $\lambda_{c2} = \frac{1}{2}$ exact: $B_P/N = 0.000000000000$ at $\lambda = 0.5$.
- λ_i strictly increasing: $0.336 < 0.345 < \cdots < 0.479$.
- B_P/A_P strictly decreasing: $1.000 \to 0.000$.
- Discrete $\lambda_{c1}(12) \approx 0.336$ vs theory $\frac{1}{3} + \frac{1}{36 \cdot 144} \approx 0.33356$. The discrete value is slightly above $\frac{1}{3}$, consistent with the theory predicting $\lambda_{c1}(K) > \frac{1}{3}$ for all finite K.

11 What Is General and What Is Specific

General (any C satisfying (C1)–(C3)): three-phase structure, $\lambda_{c2} = 1/C'(0)$, Theorem 3.1.

Specific to $C = \sin(2\, d_{\text{axis}})$, target $(1,1)$: hyperbola $A(A+B) = 4$, switching function $\lambda^*(A) = A^2/(A^2+4)$, optimal cost $d^* = 4\sqrt{\lambda(1-\lambda)}$, exact phases $\frac{1}{3}$ and $\frac{1}{2}$, convergence rate $\Theta(\Omega_K) = \frac{1}{36K^2}$.

12 Conclusion

1. The staircase paradox dissolves: $\Omega_1 \subsetneq \Omega_2$.
2. Parametric shortest-path selection is linear in (A, B)-space; phase transitions are convex-hull vertex switches (Theorem 3.1).
3. For any C satisfying (C1)–(C3): $\lambda_{c2} = 1/C'(0)$ (Theorem 4.1).
4. The true continuous frontier is the hyperbola $A(A + B) = 4$ (Theorem 5.1).
5. All key quantities are in exact closed form: $\lambda_{c1} = \frac{1}{3}$, $\lambda_{c2} = \frac{1}{2}$, $d^* = 4\sqrt{\lambda(1-\lambda)}$ (Theorems 6.3).
6. Every admissible path satisfies $A_P(A_P + B_P) \geq 4$ (proved via the identity $1 + C(\theta) = (\cos\theta + \sin\theta)^2$ and Jensen's inequality), making $A(A + B) = 4$ the global lower envelope (Theorem 5.2).
7. Global optimality follows as a corollary; the symmetric pair is also a strict local minimum with $H = kI$ (Theorem 5.3).
8. The discrete-to-continuous convergence rate is $\lambda_{c1}(K) = \frac{1}{3} + \frac{1}{36K^2} + O(K^{-3})$, with $\Theta(\Omega_K) = \frac{1}{36K^2}$ as the exact structure upgrade cost (Theorem 9.1).
9. The φ-parametrisation $\lambda = \sin^2\varphi$ gives $d^* = 2\sin 2\varphi$, $A^* = 2\tan\varphi$; the involution σ becomes $\varphi \mapsto \pi/2 - \varphi$ (Proposition 6.1).
10. n-dimensional generalisation: $A_P(A_P + B_P) \geq 2n$ for the cyclic-shift cost $C(\mathbf{e}) = \mathbf{e} \cdot \sigma(\mathbf{e})$ toward $\mathbf{1}_n$; the frontier is non-degenerate iff n is even, by Z_n eigenvalue structure (Theorems 14.1–14.2).
11. Unit-circle geometry: $(2\lambda - 1,\ d^*/2)$ lies on the unit circle; λ_{c1} corresponds to the tetrahedral angle $\arccos(-1/3) \approx 109.47°$; new product identity $d^* \cdot d^*_{\text{dual}} = 16(1-\lambda)$ (Theorem 6.6).
12. The Hessian eigenvalue $k = 2(3\lambda - 1)\sqrt{\lambda/(1-\lambda)}$ proves $k > 0$ for $\lambda \in (\frac{1}{3}, \frac{1}{2})$, closing the last open problem (Corollary 6.1).
13. Complete duality $\lambda \cdot \mu = 1$: same frontier, inverted phases, $d^*/d^*_{\text{dual}} = \lambda$, $k_{\text{primal}} = k_{\text{dual}}$ (Theorem 7.1).
14. For general $(1, r)$: $\lambda_{c1}(r) = (1 + r^2)/(6r)$, maximised mixed-phase width $\frac{1}{6}$ at $r = 1$ (Theorem 8.1).
15. The switching threshold $\lambda_{\text{sw}}(\theta)$ is strictly decreasing on $(0, \pi/4)$ because its derivative's numerator factors as $-2(\cos\theta - \sin\theta) \cdot \sqrt{2}(\cos\theta + \sin\theta - \sqrt{2})^2$ (Theorem 9.2). This closes the previously open question of which direction $(K, K-1)$ is first.
16. Phase classification by (a, b): four types, three-phase iff $0 < a < |b|$; $C = \sin(2\theta)$ is the unique algebraic-frontier cost (Theorems 13.1–13.2).

Open problems.

- Prove the convergence rate Theorem 9.1 for general C and target $(1, r)$ beyond $r = 1$.
- The constant $\frac{1}{36}$ in $\Theta(\Omega_K) = \frac{1}{36K^2}$: find a variational or geometric interpretation.
- Characterise all cost functions C for which the efficient frontier is algebraic (e.g. a conic section).

13 Phase Classification by Cost Function

The three-phase structure of Section 4 is not universal: it depends critically on two local properties of C.

Theorem 13.1 (Phase classification)**.** *Let $C\colon [0, \pi/2] \to \mathbb{R}$ satisfy* (C1)–(C3)*, be C^2, and satisfy $C(\theta) = C(\pi/2 - \theta)$ (symmetric about $\pi/4$). Define*

$$a := C'(0), \qquad b := C''(\tfrac{\pi}{4}) + C(\tfrac{\pi}{4}).$$

The parameter b equals $L^{-1} \frac{d^2}{d\phi^2} B(\phi)\big|_{\phi=0}$, so $b < 0$ means the diagonal is a local maximum of B along the symmetric family, and $b \geq 0$ means a local minimum.

The phase structure of $\min(A + \lambda B)$ *falls into one of four types:*

Type I *($b \geq 0$): diagonal optimal for all λ.*

Type II *($a > 0$, $b \geq 0$, or $a \geq |b|$ when $b < 0$): two phases,* $\mathrm{D} \to \mathrm{A}$*, with $\lambda_{c2} = 1/a$.*

Type III *($a = 0$, $b < 0$): two phases,* $\mathrm{D} \to \mathrm{M}$*, with $\lambda_{c1} = -1/b$.*

Type IV *($a > 0$, $b < 0$, $a < |b|$): three phases,* $\mathrm{D} \to \mathrm{M} \to \mathrm{A}$*, with $\lambda_{c1} = -1/b$ and $\lambda_{c2} = 1/a$.*

The three-phase condition is compactly $0 < a < |b|$.

Proof. Claim 1 ($\lambda_{c2} = 1/a$). Near the axial direction $\theta \to 0$: $C(\theta) \approx a\theta$, so $B \approx a\theta A_0$ for the axial family. The first-order condition for the axial path to become locally optimal gives $\lambda_{c2} = 1/a$. If $a = 0$, the penalty vanishes to second order at the axial, and $\lambda_{c2} = \infty$: the axial phase never appears.

Claim 2 ($\lambda_{c1} = -1/b$). At $\phi = 0$ (diagonal), $A''(0) = L$ and $B''(0) = L\big(C''(\pi/4) + C(\pi/4)\big) = Lb$. The second-order stationarity condition $A''(0) + \lambda\, B''(0) = 0$ gives $\lambda_{c1} = -1/b$. If $b \geq 0$, then $B''(0) \geq 0$: the diagonal is a local minimum of B, so the condition is never satisfied at finite λ, and $\lambda_{c1} = \infty$: the diagonal–interior transition does not occur.

Ordering. $\lambda_{c1} < \lambda_{c2}$ iff $-1/b < 1/a$ iff $a < |b|$ (for $a, b > 0, b < 0$). When $a \geq |b|$, both transitions exist but $\lambda_{c1} \geq \lambda_{c2}$: the mixed phase collapses and the system goes $\mathrm{D} \to \mathrm{A}$. □

Remark 13.1 (Generic behaviour)**.** *For any smooth symmetric C with a strict maximum at $\theta = \pi/4$, we have $C''(\pi/4) < 0$, so $b < 0$ generically. Thus Types III and IV are the generic cases; Type I requires an unusually flat peak.*

Theorem 13.2 (Uniqueness of algebraic frontier)**.** *Among all C satisfying the hypotheses of Theorem 13.1, the function $C(\theta) = \sin(2\theta)$ is the unique one for which the frontier $A(A + B) = \mathrm{const}$ holds pointwise along the symmetric family (equivalently, the frontier is an algebraic curve on that family).*

Table 3: Phase types: verified examples with $C(\theta)$ smooth, symmetric.

Type	Conditions	Sequence	λ_{c1}	λ_{c2}	Example
I	$b \geq 0$	D only	∞	∞	(requires flat peak)
II	$a > 0,\ b \geq 0$	D→A	∞	$1/a$	—
II'	$a > 0,\ b < 0,\ a \geq \lvert b\rvert$	D→A	$> 1/a$	$1/a$	parabola $\theta(\pi/2-\theta)$
III	$a = 0,\ b < 0$	D→M	$-1/b$	∞	$\sin^2(2\theta)$
IV	$a > 0,\ b < 0,\ a < \lvert b\rvert$	D→M→A	$-1/b$	$1/a$	$\sin(2\theta)$

Proof. The constant-frontier condition along the symmetric family requires

$$\frac{1 + \frac{1}{2}(C(\alpha-\phi) + C(\alpha+\phi))}{\cos^2\phi} = c_0 \quad \text{for all } \phi \in [0, \alpha).$$

For $\alpha = \pi/4$, the symmetry $C(\theta) = C(\pi/2 - \theta)$ gives $C(\pi/4 - \phi) = C(\pi/4 + \phi)$, so the condition reduces to

$$C\big(\tfrac{\pi}{4} - \phi\big) = c_0 \cos^2\phi - 1.$$

Setting $u = \pi/4 - \phi$ and using $\cos^2(\pi/4 - u) = (1 + \sin 2u)/2$:

$$C(u) = c_0\,\frac{1 + \sin 2u}{2} - 1.$$

With $c_0 = 2$: $C(u) = \sin(2u)$. For $c_0 \neq 2$ the boundary conditions $C(0) = 0$ or $C(\pi/2) = 0$ fail. Hence $C(\theta) = \sin(2\theta)$ is the unique solution. □

Remark 13.2. *This uniqueness explains why $C = \sin(2\theta)$ yields the hyperbola $A(A + B) = 4$ with exact closed forms for all key quantities (Sections 6–7): it is the only cost function for which the Pareto frontier is algebraic rather than transcendental.*

14 High-Dimensional Generalisation

The lower-envelope inequality and the even–odd structure both generalise to arbitrary dimension with complete proofs.

Theorem 14.1 (n-dimensional lower envelope)**.** *Let $\sigma\colon \mathbb{R}^n \to \mathbb{R}^n$ be the cyclic shift $\sigma(a_1, \ldots, a_n) = (a_2, \ldots, a_n, a_1)$, and set $C(\mathbf{e}) = \mathbf{e} \cdot \sigma(\mathbf{e})$ for $\mathbf{e} \in S^{n-1}$. For any path from $\mathbf{0}$ to $\mathbf{1}_n = (1, \ldots, 1)$ using directions $\mathbf{e}_1, \ldots, \mathbf{e}_m \in S^{n-1}$ with steps $s_1, \ldots, s_m > 0$,*

$$A_P(A_P + B_P) \geq 2n,$$

with equality if and only if all $\mathbf{e}_i + \sigma(\mathbf{e}_i) \propto \mathbf{1}_n$.

Proof. Set $w_i = s_i/A_P > 0$ with $\sum_i w_i = 1$, and $\mathbf{v}_i = \mathbf{e}_i + \sigma(\mathbf{e}_i)$.

Step 1 (identity). Since $|\mathbf{e}_i| = |\sigma(\mathbf{e}_i)| = 1$,

$$|\mathbf{v}_i|^2 = 2 + 2\mathbf{e}_i \cdot \sigma(\mathbf{e}_i) = 2(1 + C(\mathbf{e}_i)).$$

Using $B_P = \sum_i s_i C(\mathbf{e}_i) = A_P \sum_i w_i C(\mathbf{e}_i)$:

$$A_P(A_P + B_P) = A_P^2\left(1 + \sum_i w_i C(\mathbf{e}_i)\right) = A_P^2 \sum_i w_i(1 + C(\mathbf{e}_i)) = \frac{A_P^2}{2}\sum_i w_i |\mathbf{v}_i|^2.$$

Step 2 (feasibility). $\sigma(\mathbf{1}_n) = \mathbf{1}_n$, so $\sum_i w_i\sigma(\mathbf{e}_i) = \sigma(\sum_i w_i\mathbf{e}_i) = \mathbf{1}_n/A_P$. Hence $\sum_i w_i\mathbf{v}_i = (2/A_P)\mathbf{1}_n$ and $|\sum_i w_i\mathbf{v}_i|^2 = 4n/A_P^2$.

Step 3 (Jensen). The map $\mathbf{v} \mapsto |\mathbf{v}|^2$ is convex, so

$$\sum_i w_i|\mathbf{v}_i|^2 \;\geq\; \Big|\sum_i w_i\mathbf{v}_i\Big|^2 = \frac{4n}{A_P^2}.$$

Combining Steps 1–3: $A_P(A_P + B_P) \geq \frac{A_P^2}{2} \cdot \frac{4n}{A_P^2} = 2n$.

Equality. Jensen's inequality for $|\cdot|^2$ is tight iff all $\mathbf{v}_i$ are parallel (equal direction). Since each $\mathbf{v}_i = \mathbf{e}_i + \sigma(\mathbf{e}_i)$, this is exactly $\mathbf{e}_i + \sigma(\mathbf{e}_i) \propto \mathbf{1}_n$. □

Theorem 14.2 (Even–odd parity splitting)**.** *For the cyclic-shift problem with target $\mathbf{1}_n$:*

(i) Even $n = 2k$*: the* alternating family $\mathbf{e} = (a, b, a, b, \ldots, a, b)$ *with $a, b > 0$ and $2k(a^2 + b^2) = 1$ satisfies $\mathbf{e} + \sigma(\mathbf{e}) = (a + b)\mathbf{1}_n$, so every* single *alternating direction achieves equality in Theorem 14.1 when used alone. Two alternating directions with the same value of $(a+b)$ also achieve equality in combination; directions with different $(a + b)$ values do not, since $|\cdot|^2$ is strictly convex and parallel-but-unequal $\mathbf{v}_i$ cannot satisfy Jensen with equality.*

(ii) Odd $n = 2k + 1$*: the only direction satisfying $\mathbf{e} + \sigma(\mathbf{e}) \propto \mathbf{1}_n$ is the diagonal $\mathbf{e} = \mathbf{1}_n/\sqrt{n}$. The frontier consists of a single point.*

Proof. Write $\mathbf{e} = (a_1, \ldots, a_n)$. The condition $\mathbf{e} + \sigma(\mathbf{e}) \propto \mathbf{1}_n$ means $a_i + a_{i+1} = c$ for all i (indices mod n), for some constant $c > 0$.

(i) Even $n = 2k$. The alternating vector $(a, b, a, b, \ldots, a, b)$ satisfies $a+b = b+a = \cdots = c$ for any $a, b > 0$, giving a one-parameter family. Conversely, subtracting consecutive equations $a_i + a_{i+1} = c$ and $a_{i+1} + a_{i+2} = c$ gives $a_i = a_{i+2}$ for all i, so $a_1 = a_3 = a_5 = \cdots$ and $a_2 = a_4 = a_6 = \cdots$. For even n these two subsequences are independent, yielding the alternating family.

(ii) Odd $n = 2k + 1$. The same argument gives $a_1 = a_3 = \cdots = a_{2k+1} = a_2$ (since the odd-indexed subsequence wraps around to include a_{2k+1} and then $a_1 = a_3$ forces $a_{2k+1} = a_2$). Hence all a_i are equal, so $\mathbf{e} = \mathbf{1}_n/\sqrt{n}$.

Algebraic reason. The cyclic permutation matrix has eigenvalues $\omega^k = e^{2\pi ik/n}$, $k = 0, \ldots, n-1$. The alternating condition corresponds to the eigenvalue -1, which exists iff n is even (since $e^{2\pi ik/n} = -1$ requires $n \mid 2k$ with $k = n/2$, possible only for even n). □

Corollary 14.1 (n-dimensional global optimality)**.** *For any $\lambda \geq 0$, the minimum of $A_P + \lambda B_P$ over all admissible paths equals $d_n^*(\lambda)$, achieved by paths whose directions satisfy $\mathbf{e}_i + \sigma(\mathbf{e}_i) \propto \mathbf{1}_n$.*

Proof. The set $\{(A, B) : A(A + B) \geq 2n,\ A, B \geq 0\}$ is a closed convex epigraph, and by Theorem 14.1 every feasible path (A_P, B_P) lies in this set. The linear objective $A + \lambda B$ is minimised on the epigraph boundary $A(A + B) = 2n$, at the directions achieving equality. □

Theorem 14.3 (Scaling equivalence for even n)**.** *Let $n = 2k$. Parametrise the alternating family by $\mathbf{e} = (\cos\theta/\sqrt{k}, \sin\theta/\sqrt{k}, \ldots)$ with $\theta \in (0, \pi/2)$. Set $\tilde{A} = A/\sqrt{k}$ and $\tilde{B} = B/\sqrt{k}$. Then $C(\mathbf{e}) = \sin(2\theta)$ for all k, and $(\tilde{A}, \tilde{B})$ satisfies*

$$\tilde{A}(\tilde{A} + \tilde{B}) = 4,$$

the same frontier equation as $n = 2$. Consequently:

(i) $d_n^*(\lambda) = \sqrt{k}\, d_2^*(\lambda)$ for all $\lambda \geq 0$.

(ii) The phase boundaries are universal: $\lambda_{c1} = \frac{1}{3}$ and $\lambda_{c2} = \frac{1}{2}$ for every even n.

(iii) $d_n^* \cdot d_{n,\text{dual}}^* = 8n(1-\lambda)$.

(iv) $(2\lambda-1)^2 + (d_n^*/(2\sqrt{k}))^2 = 1$ *(unit-circle representation, same circle for all even n).*

Proof. Direct computation: $C(\mathbf{e}) = \sum_{i=1}^{n} e_i e_{i+1} = k(ab+ba) = 2k\cdot(\cos\theta/\sqrt{k})(\sin\theta/\sqrt{k}) = \sin(2\theta)$. The path uses $\mathbf{e}$ and $\sigma(\mathbf{e})$ each for $t = \sqrt{k}/(\cos\theta+\sin\theta)$ steps, giving $A = 2\sqrt{k}/(\cos\theta+\sin\theta) = \sqrt{k}\,\tilde{A}$ where $\tilde{A} = 2/(\cos\theta+\sin\theta)$ is the 2D quantity. Similarly $B = \sqrt{k}\,\tilde{B}$. Then $\tilde{A}(\tilde{A}+\tilde{B}) = 4$ by the 2D result. All four consequences follow by substituting $A = \sqrt{k}\,\tilde{A}$, $B = \sqrt{k}\,\tilde{B}$ into the 2D formulae. The lower bound $A_P(A_P+B_P) \geq 2n$ holds for all admissible paths (Theorem 14.1). The minimum over the single-direction alternating family gives $d_n^*(\lambda) = \sqrt{k}\,d_2^*(\lambda)$, which is a global lower bound; whether a non-alternating path can undercut it remains open. □

Corollary 14.2 (Universal phase boundaries)**.** *For every even n, the phase boundaries of the n-dimensional cyclic-shift problem restricted to the alternating family are $\lambda_{c1} = \frac{1}{3}$ and $\lambda_{c2} = \frac{1}{2}$, independent of n.*

Remark 14.1 (Autocorrelation structure)**.** *For the alternating family with parameter θ, the Define the lag-j inner product $C_j(\mathbf{e}) = \mathbf{e}\cdot\sigma^j(\mathbf{e})$ for $j = 0,\ldots,n-1$. For the alternating family it takes exactly two values:*

$$C_j = \begin{cases} 1 & j \text{ even}, \\ \sin(2\theta) & j \text{ odd}. \end{cases}$$

This two-valued structure holds for all even n and all $\theta \in (0,\pi/2)$.

Remark 14.2 (Two-dimensional richness)**.** *The dimension $n = 2$ is the smallest even integer, hence the simplest case with a non-trivial symmetric family. By Theorem 14.3, every even-n problem is a $\sqrt{k}$-rescaling of the 2D problem on the alternating family. All the rich structure of the 2D problem—the hyperbolic frontier, universal phase transitions at $\lambda_{c1} = \frac{1}{3}$ and $\lambda_{c2} = \frac{1}{2}$, the φ-parametrisation, duality, unit-circle geometry—transfers exactly to every even n via this scaling. For odd n the frontier degenerates to a single point.*

Open Problems

This section records personal intuitions and cross-domain analogies. These are not mathematical claims; no proofs are offered or implied. The author does not assert that any of the connections below are rigorous.

Pareto frontier and the structure of trade-offs. The hyperbola $A(A+B) = 4$ is a Pareto frontier: no path simultaneously minimises both A and B. The primal–dual correspondence $\lambda\cdot\mu = 1$ says that the "shadow price" of each objective is the reciprocal of the other. Similar trade-off curves appear in economics (production possibility frontiers), information theory (rate–distortion), and thermodynamics (efficiency vs. power). Whether these share a deeper common structure beyond the formal analogy is an open question.

Phase transitions and structure upgrade. The discrete convergence $\Theta(\Omega_K) = 1/(36K^2) \to 0$ captures the cost of using a coarser direction vocabulary. This looks structurally like the cost of a "model class" in minimum description length theory, where richer models pay a description cost. Whether the specific constant $\frac{1}{36}$ has a universal interpretation (e.g. as a Fisher information or a curvature term) is unknown to the author.

Involution and symmetry breaking. The involution $\sigma : A \mapsto 4/A$ leaves d^* invariant while complementing $\lambda \mapsto 1 - \lambda$. This is reminiscent of Kramers–Wannier duality in the 2D Ising model, where a high-temperature/low-temperature duality fixes the critical point. In our problem the "self-dual" point is $A = 2$ ($\lambda_{c2} = \frac{1}{2}$), which is the upper phase boundary. Whether this analogy can be made precise is unclear.